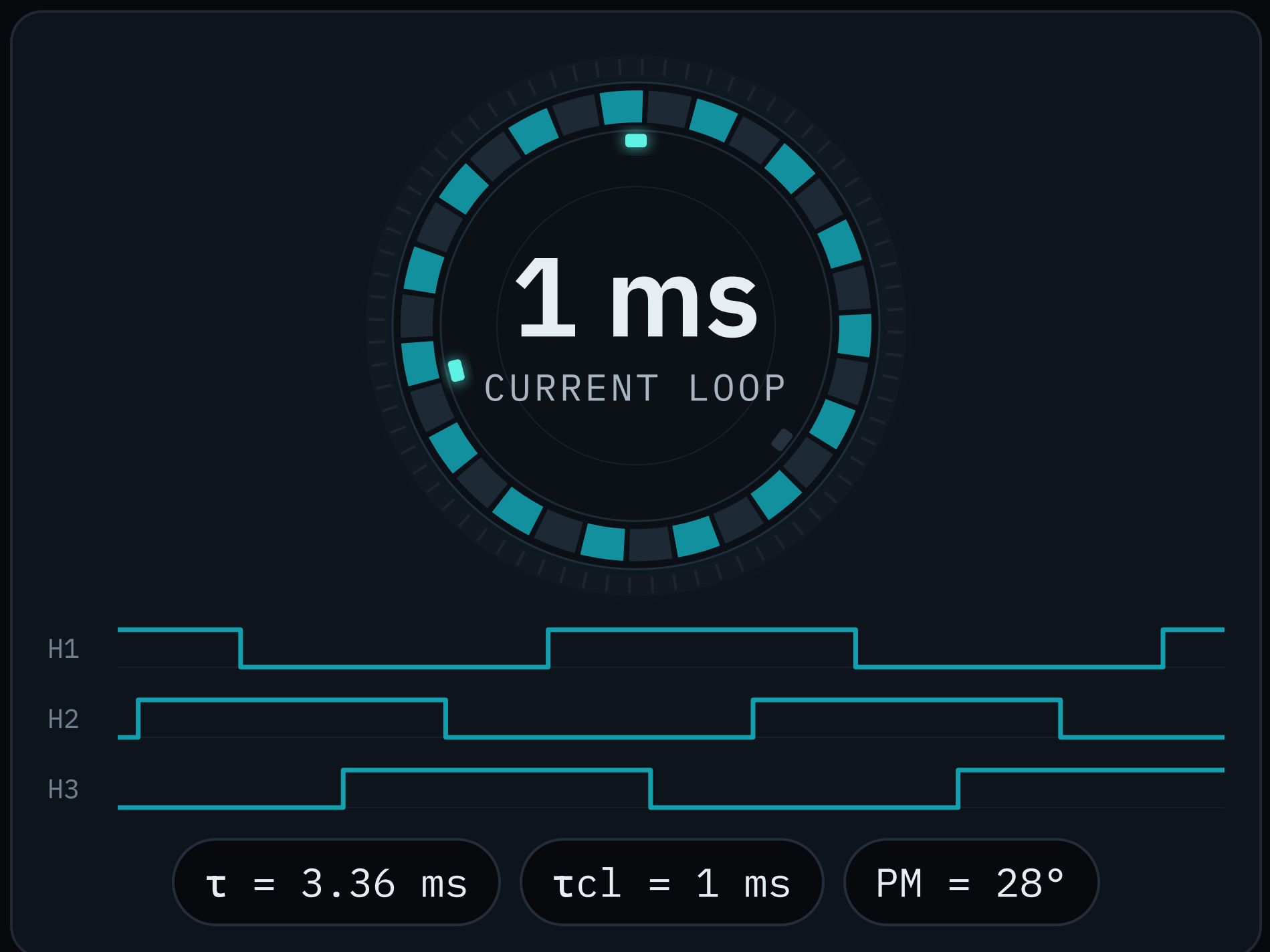


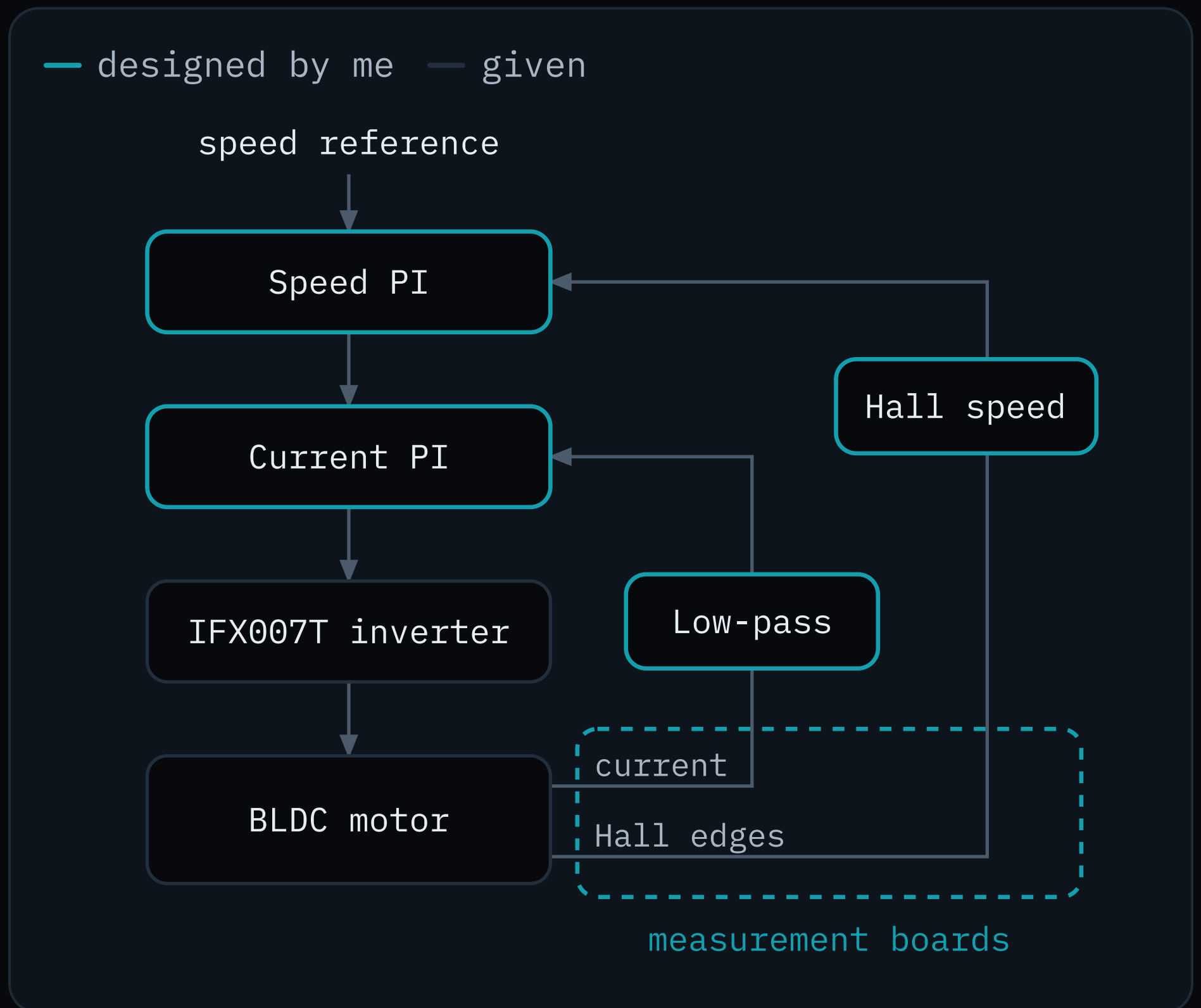
Controlling a BLDC hub motor

Motor characterisation, current loop and speed loop on an STM32. My part of a team project.



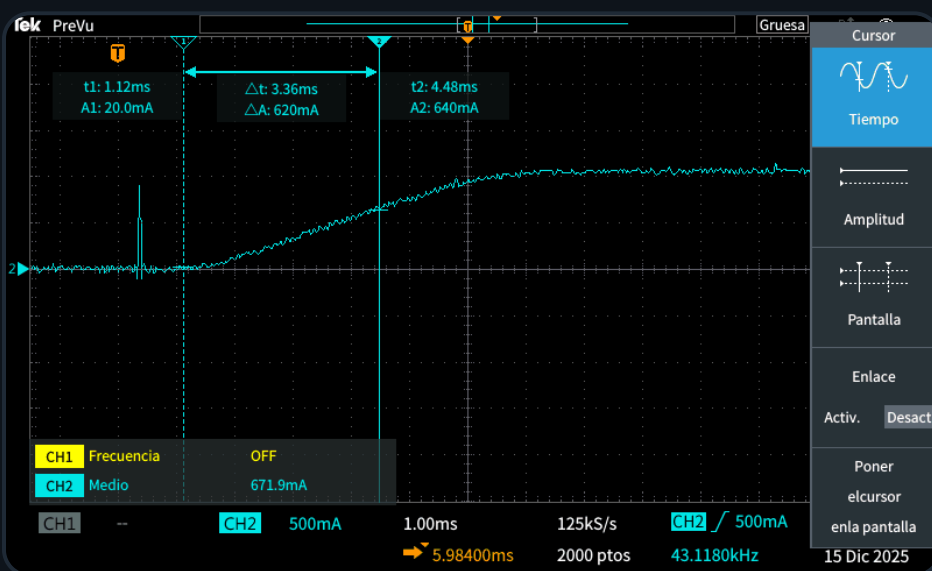
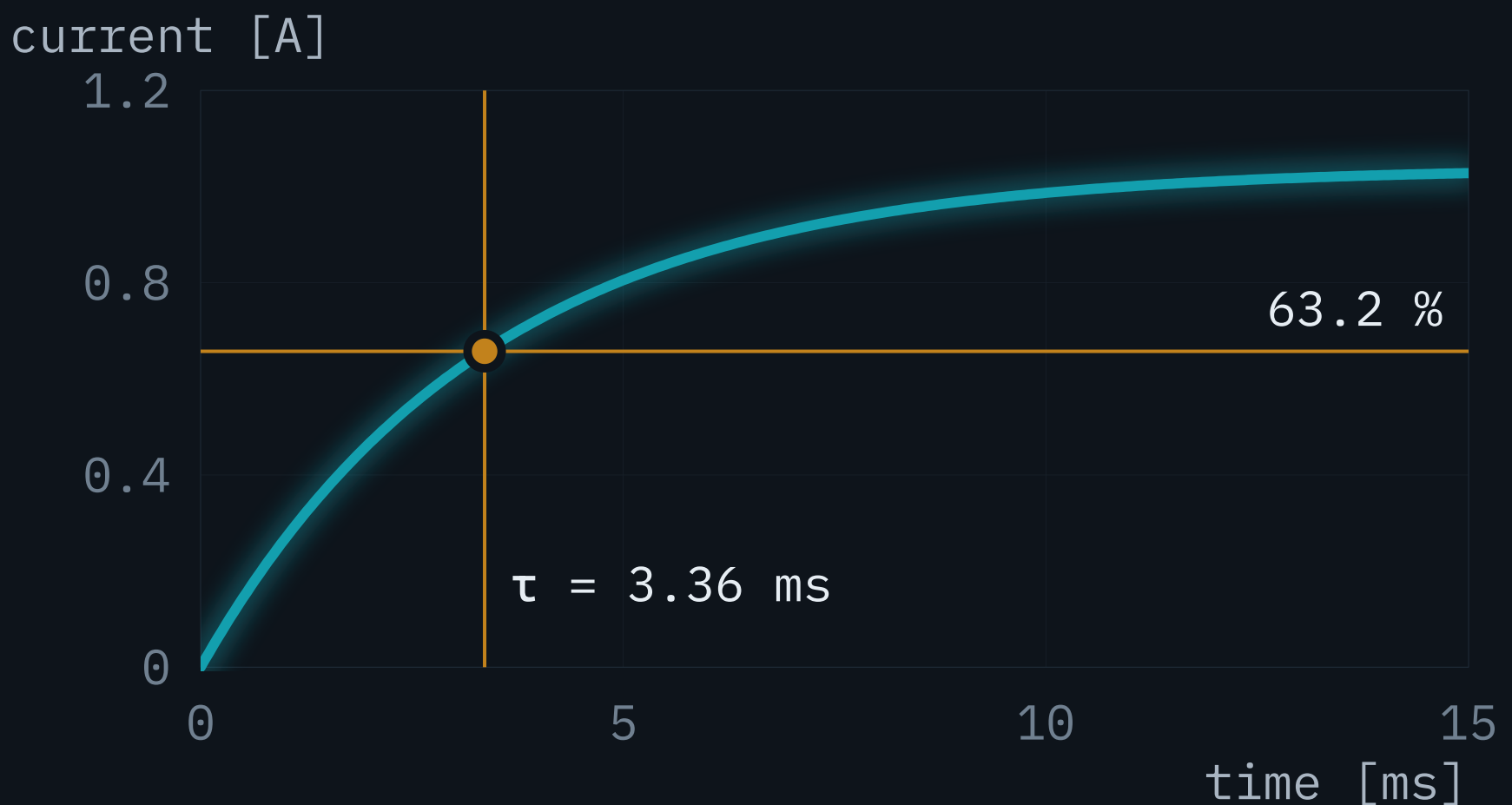
Scope of my work

The motor and the IFX007T inverter were given. I designed and sized the **measurement boards**, minimising electromagnetic noise, and the **control on the STM32**.



Motor characterisation

Locked rotor and a voltage step: the current reaches 63.2 % after $\tau = 3.36 \text{ ms}$. The Simulink motor model gives 3.37 ms.



Oscilloscope capture,
15 Dec 2025.

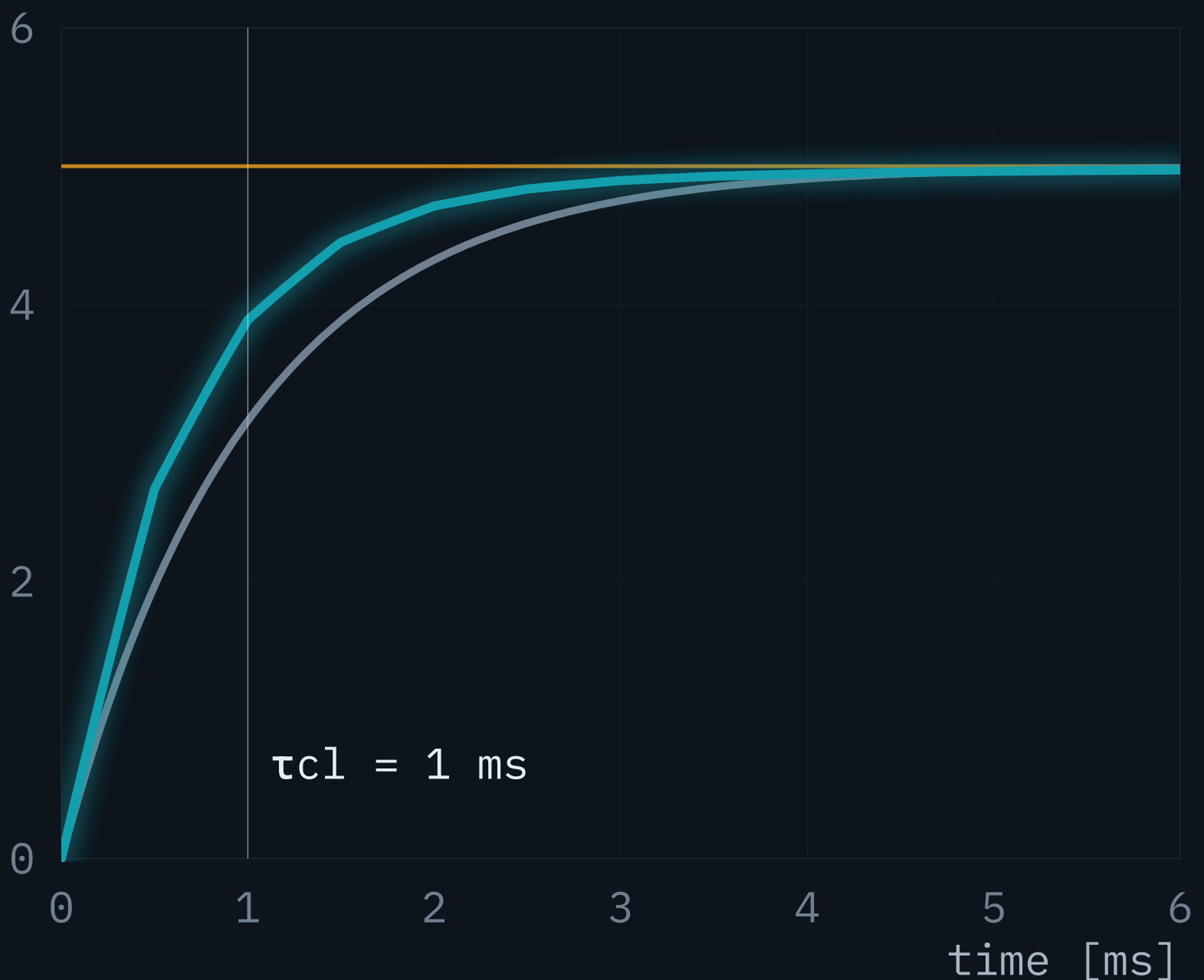
$\Delta t = 3.36 \text{ ms}$ from the
step to 63.2 % of the
final current.

Current loop: $\tau_{cl} = 1 \text{ ms}$

With $T_i = L/R$ the PI zero cancels the motor pole. $K_p = L/\tau_{cl} = 1.06$ gives a first-order loop without overshoot.

— $I^* = 5 \text{ A}$ — continuous — sampled 0.5 ms

current [A]

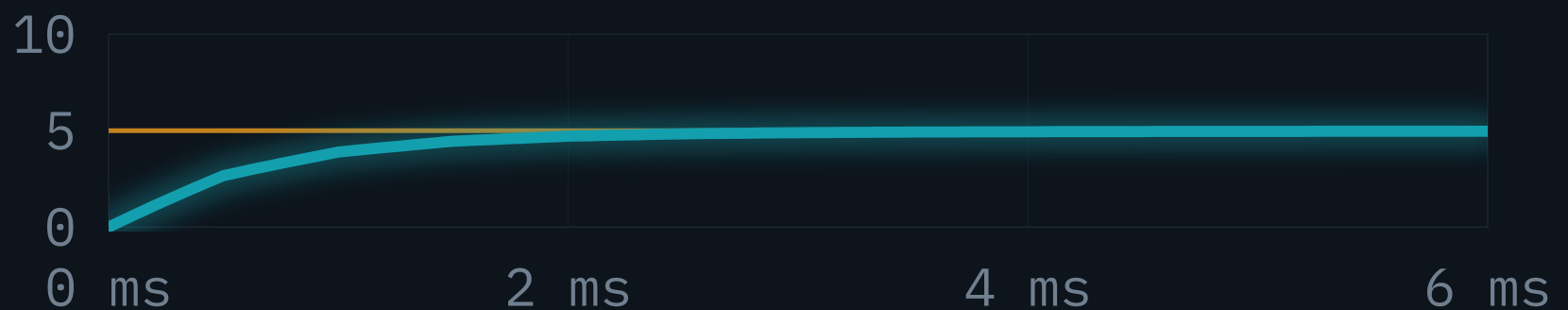


Limit of the 0.5 ms sampling

The controller runs every 0.5 ms. Above $K_p \approx 2$ the loop overshoots, and above $K_p \approx 3.9$ it is unstable.

$K_p = 1.06$

no overshoot



$K_p = 3$

51 % overshoot



$K_p = 4.5$

unstable



Current filter and phase margin

A 250 rad/s low-pass filter reduces the noise of the Hall current sensors. The phase margin goes from **90° to 28°**.



Speed from the Hall sensors

No encoder: 15 pole pairs $\times$ 6 edges = **90 edges per turn**. At each edge, $\omega = (2\pi/90)/\Delta t$.



90

edges per turn

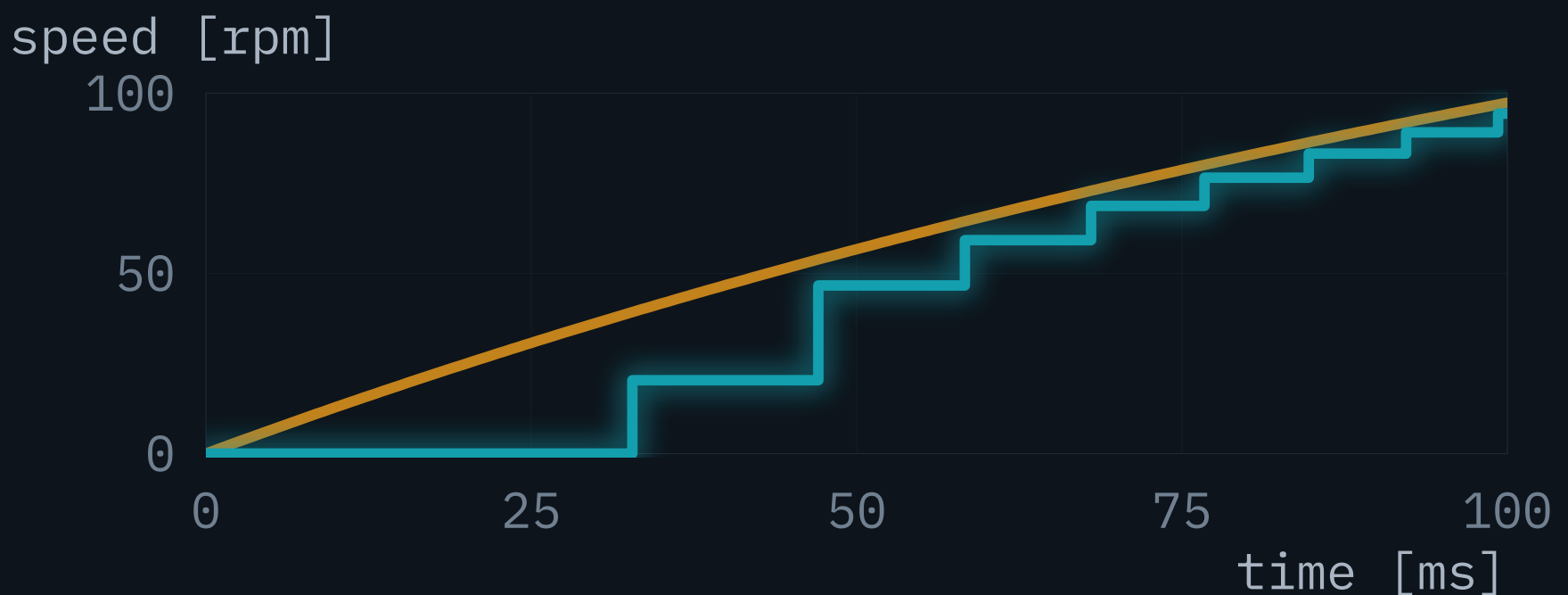
300 /s

edges at 200 rpm

3.3 ms

between new values

— actual speed — Hall estimate

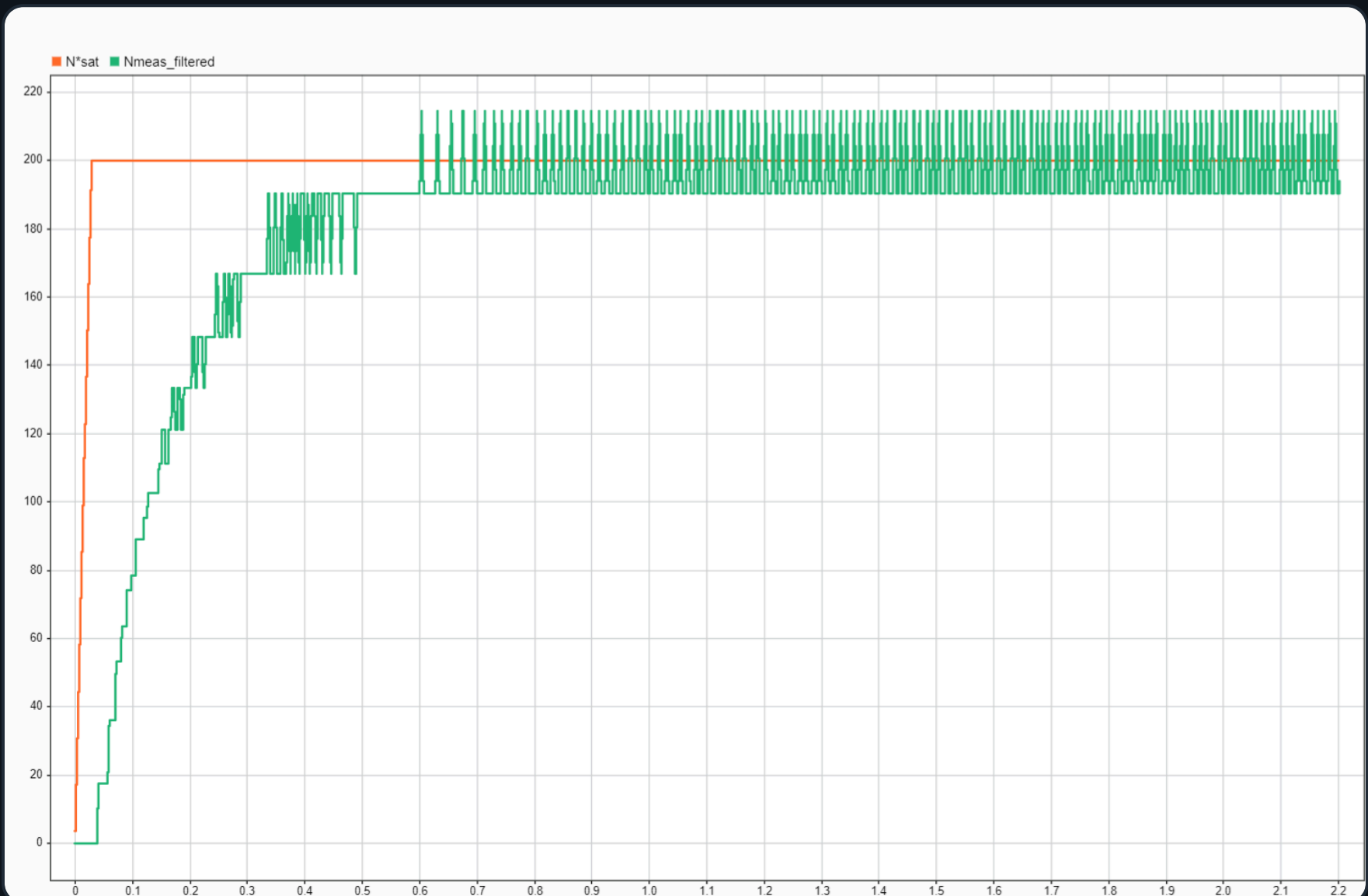


Speed loop

PI with acceleration feed-forward and anti-windup, on top of the current loop. In Simulink it reaches **200 rpm in about 0.6 s.**

$$K_p = 0.0177 \text{ A/rpm}$$

$$T_i = 2.0 \text{ s}$$



Simulink, complete drive: N^* (orange) and the Hall-based speed (green), in rpm.

Results

The current follows a 5 A step in a few milliseconds. On the scooter, the drive could be **driven, accelerated and braked** correctly.



Simulink, complete drive: I^* (purple) and the measured current (green), in A. The dips are the commutations.

The drive in five numbers

- **3.36 ms** motor time constant, measured
- **1 ms** current loop, $K_p = 1.06$
- **28°** phase margin with the 250 rad/s filter
- **90** Hall edges per turn
- **0.6 s** to 200 rpm in Simulink

INTERACTIVE VERSION

perumendivil.github.io/blcdc-drive

I'm looking for a Bachelor's thesis in automation or robotics in Västra Götaland, January-June 2027.

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